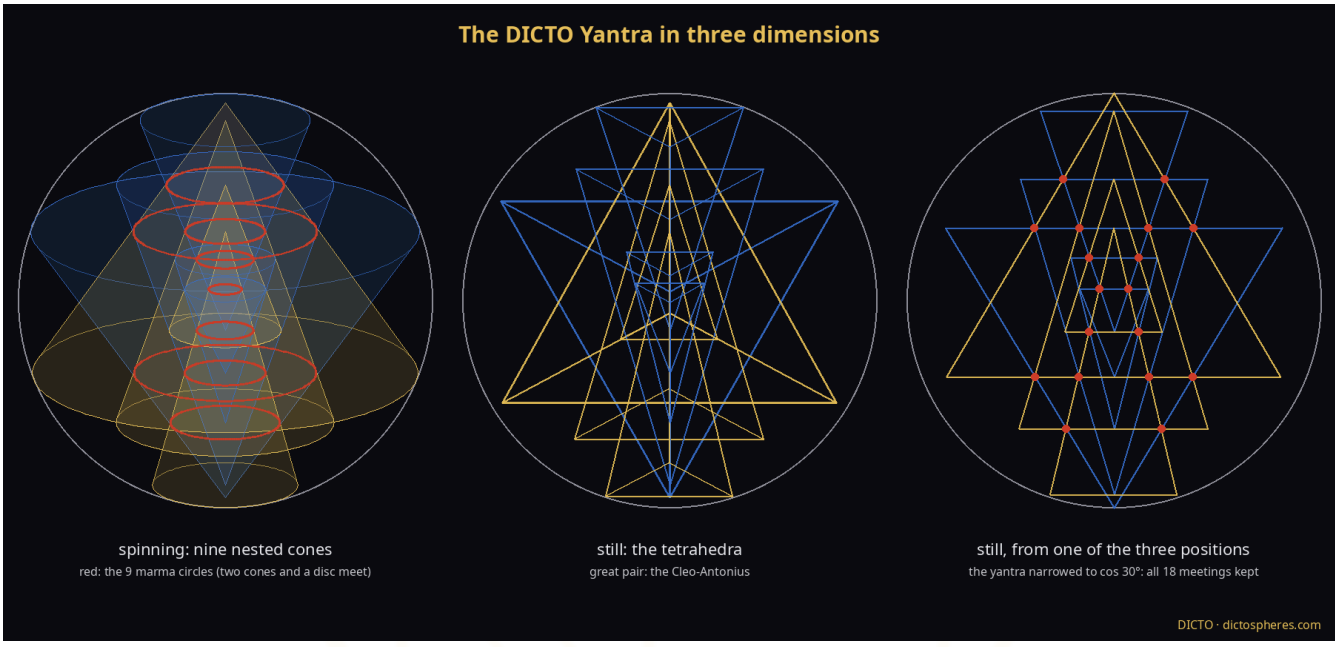
The DICTO Yantra

Exact Śrī Yantras, a Rational One, and a Perfect Yantra on the Cleo-Antonius

DICTO (artist and geometer, Osaka, Japan) · dictospheres.com · contact@dictospheres.com · ORCID 0009-0001-2620-2036
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The rhombus, the solids, the viewpoint, the Cleo-Antonius and the names are DICTO's. Computation and drafting with Claude (Anthropic).



A Śrī Yantra is nine triangles in a circle, four pointing up and five pointing down, drawn so that their sides cross three at a time at eighteen points. Drawn by hand, the crossings never quite close; the figure has been drawn by eye and solved by numbers. This paper shows exact ones: a yantra whose every corner is a fraction, yantras built on solids that DICTO proposed, and the DICTO Yantra, which meets every traditional demand at once. It ends with a choice every exact yantra has to make.

Abstract. The Śrī Yantra asks its nine triangles to meet three at a time at 18 points (the marmas), to make 43 small triangles, and, in the strict reading, to touch the circle with its smallest top and bottom triangles and to centre the innermost triangle (the bindu). We prove that every rational choice of four natural parameters gives a yantra with all 18 meetings exact in one quadratic field (Theorem 1), and give a yantra with every coordinate a fraction (Theorem 2). The 43 are the pieces lying inside an odd number of the nine triangles (Proposition 3). Spun about its axis the yantra is the axial section of nine nested cones; still tetrahedra show it from three positions narrowed to cos 30° (Proposition 4). The perspectives that keep the circle round keep every meeting and the product of the two great slopes (Theorem 5); in natural coordinates the yantra is one quadratic and one relation of degree 10 (Theorem 6). With one slope for all nine triangles

the heights are forced to 2 : 4 : 4 : 4 : 6 : 6 : 8 : 8 : 9 and the great triangles cannot both be inscribed (Theorem 7). Computed to 60 digits, with exact forms open: the **DICTO Yantra**, on DICTO's Cleo-Antonius pair (slope product $2/\varphi^3$), meets every condition, with 43 triangles, 18 marmas and 24 sandhis; it is the Cleo-Antonius seen from one viewpoint. The most even exact yantra keeps the 43 but leaves the circle. Every exact yantra found chooses between its small triangles touching the circle and its pieces being even: the DICTO Yantra is perfect by the rules, the most even yantra perfect to the eye.

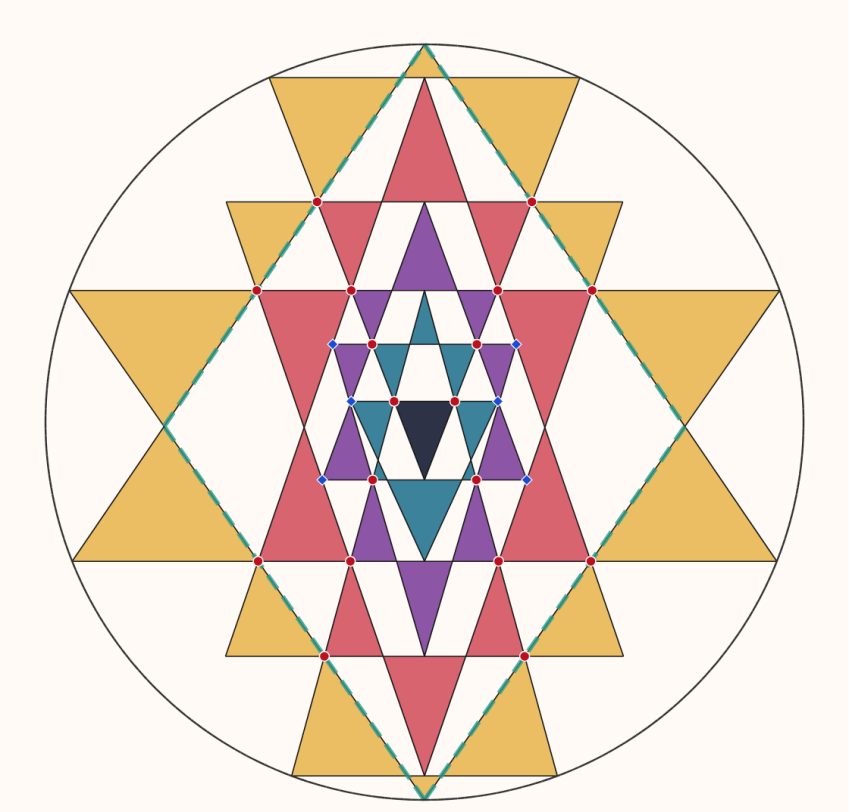


Figure 1. The DICTO Yantra. Great pair: the Cleo-Antonius seen from the viewpoint that centres the bindu (slopes 0.6793 above, 0.6950 below, product exactly $2/\varphi^3$). Both small triangles touch the circle, the bindu is centred, 43 triangles (shaded ring by ring), 18 marmas (red), the six corner touches (blue), DICTO's rhombus (dashed; here a kite).

1. The yantra and its demands

Model. The circle has diameter 1; depth d is measured down from its top point O, T is its bottom point. The nine triangles t1 … t9 follow Chiodo's numbering, bases from top to bottom: t1–t5 point down, t6–t9 point up. Triangle ti has base depth b_i, half-width w_i and apex depth a_i. The two great triangles are inscribed: t7 has its apex at O, t3 at T, and their corners lie on the circle. The upper great triangle t7 has leg slope u (half-width per depth from O), the lower one t3 has leg slope v (from T). Writing $P = b_3$, $Q = b_6$, $R = b_7$, $S = b_9$ (Chiodo's parameters), the circle gives $P = v^2/(1+v^2)$, $R = 1/(1+u^2)$, $w_3 = v/(1+v^2)$, $w_7 = u/(1+u^2)$: the rational parametrisation of the circle.

The demands. (i) Apex on base: $a_8 = b_1$, $a_6 = b_2$, $a_9 = b_3$, $a_1 = b_6$, $a_5 = b_7$, $a_4 = b_8$, $a_2 = b_9$. (ii) The **18 marmas**, nine mirror pairs of points where a down-leg, a base and an up-leg meet: (1,2,7), (2,3,7), (1,3,8), (1,4,6), (1,5,9), (4,6,9), (2,7,9), (3,7,8), (3,8,9). Their incidences per triangle are 8, 6, 8, 4, 2, 4, 8, 6, 8, as in Kavi Mahesh's table. (iii) Six **corner touches**, a

corner of one triangle on another's side: (4,4,8), (5,5,6), (2,6,6). These are the conditions of Chiodo, equivalent to Huet's; every yantra in this paper satisfies them all. Kulaichev, Rao and Pustynski add four more: (a) t3 ≅ t7; (b) t1's base corners on the circle; (c) t9's base corners on the circle; (d) the incircle of the bindu (the innermost triangle, between t1's legs and t5's base) centred on the circle's centre. Tradition also counts 43 small triangles and 24 **sandhis** (points where exactly two sides cross). We call a yantra **perfect** when it meets (i)–(iii), (b), (c), (d) and has 43 triangles, 18 marmas and 24 sandhis.

Reference images. DICTO's reference was a photograph of a gold embossed yantra (338 px, tilted): its layout matches the model (t7's apex at the top, t3's at the bottom, nine level lines, 4 + 5 triangles); a rough reading gives P ≈ 0.39, R ≈ 0.63, u ≈ v ≈ 0.8, not precise enough to tell a kite from a rhombus. Later DICTO compared the exact yantras with H. J. T. Matos's painted yantra (2023, CC BY-SA 4.0), used as a reference only and not reproduced here.

2. Exact yantras

> **Theorem 1 (the quadratic family).** If u, v, Q and S are rational, every corner and every meeting point of the yantra lies in one quadratic field $\mathbb{Q}(\sqrt{\Delta})$, and all 18 marmas, the 6 corner touches, the 7 apex-on-base points and both inscribed triangles hold exactly. Here Δ(u, v, Q, S) is the discriminant of the construction's closing quadratic; its square-free part has degrees 10, 20, 6, 6 in u, v, Q, S.

Proof. Given u and v, the great triangles are rational (above). Following Chiodo's construction (his Fig. 4), every further line is found by intersecting lines already drawn, a rational operation, except one step: the closing step (Apollonius' circle–line–point problem) is a quadratic equation in b_1. All later coordinates are rational in b_1 and the inputs, so they lie in $\mathbb{Q}(\sqrt{\Delta})$. The script quadratic_family.py re-proves all 23 condition expressions with exact fractions in $\mathbb{Q}(\sqrt{\Delta})$, without floating point, for the examples of Table 1. ∎

yantra	u, v, Q, S	field	max |Δb| from Huet's classical depths	odd pieces by ring
R1	19/24, 2/3, 8/15, 8/9	$\mathbb{Q}$ (Δ a square)	0.054	14, 10, 10, 8, 1
Q1	13/16, 12/17, 29/54, 5/6	$\mathbb{Q}(\sqrt{\Delta})$	0.0031	14, 10, 10, 8, 1
Q2	4/5, 7/10, 8/15, 5/6	$\mathbb{Q}(\sqrt{\Delta})$	0.020	14, 10, 10, 8, 1
Q3	3/4, 3/4, 11/20, 11/12 (3-4-5 rhombus)	$\mathbb{Q}(\sqrt{\Delta})$	0.082	14, 10, 12, 8, 1

Table 1. Examples of Theorem 1 (study data quadratic-examples.json, quadratic-family-data.txt; factor list of Δ in discriminant-log-data.txt).

> **Theorem 2 (a rational Śrī Yantra).** The yantra R1 with u = 19/24, v = 2/3, Q = 8/15, S = 8/9 has every corner and every meeting point rational. Its base depths are b = 248/3611, 48/233, 4/13, 23/61, 86/197, 8/15, 576/937, 172/229, 8/9 and its half-widths w = 836/3611, 3401/11883, 6/13, 1805/12444, 703/6698, 38/255, 456/937, 16245/50609, 323/1485. Its inscribed triangles sit on the Pythagorean triples 5-12-13 (t3) and 215-912-937 (t7).

Proof. For these values Δ is a square, so by Theorem 1 the field is $\mathbb{Q}$. The conditions were checked with exact fractions by quadratic_family.py and, independently, by check_r1_independent.py, written apart from the model: it rebuilds the nine triangles from

the stored fractions alone and finds exactly 18 points where three sides meet, with the two great triangles' corners exactly on the circle. ∎

R1 was the only rational yantra in two searches: 9.7 million parameter sets (denominators up to 40), then 627 million values of S over 14 336 slices (u, v, Q) with denominators of S up to 1200, the search re-finding R1 as a self-test. The fibres are curves of genus 2, so rational yantras are expected to be rare. R1 is admissible but higher than the traditional drawing: t1's base lies at 3.3/48 against 6/48 on the traditional grid, and its top and bottom small triangles sit inside the circle.

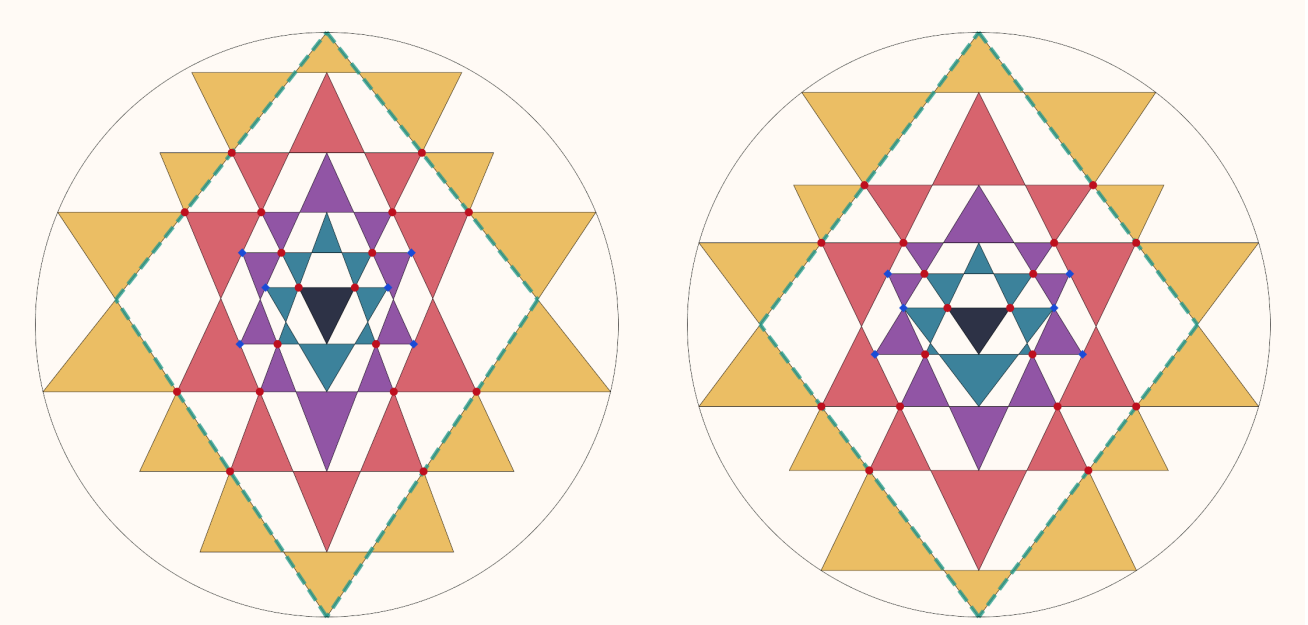


Figure 2. Left: R1, the rational Śrī Yantra (Theorem 2); its rhombus is a kite. **Right: Rh345**, on DICTO's rhombus made the 3-4-5 rhombus, exact in a field of degree 20 (§3); the two extra tiny pieces sit near the centre. Red: the 18 marmas; blue: the six corner touches; dashed: DICTO's rhombus.

> **Proposition 3 (the 43).** In every yantra of this paper with the traditional pattern (R1, Q1, Q2, K1, K2, V1, G1, PF2, the stella rhombus yantra, the DICTO Yantra, E1), the 43 small triangles of tradition are exactly the pieces of the figure that lie inside an odd number of the nine triangles: 14, 10, 10, 8 and 1 at containment depth 1, 3, 5, 7, 9. With them come 18 marmas; the sandhis number 24 in all of them except E1 (28).

Check. figure.py builds the full arrangement of the 27 sides, lists its faces, and counts for each face the triangles containing it; counts.py counts, with exact fractions from the stored 40-digit values, the points where exactly three sides meet (marmas) and exactly two (sandhis), leaving out corners. The traditional shading alternates from ring to ring, so the shaded pieces are the odd ones. ∎

3. DICTO's rhombus

DICTO proposed the upright rhombus: the two legs of t7 (from O) and the two legs of t3 (from T), the four longest lines of the yantra, meet at $E_{\pm} = (\pm uv/(u+v), v/(u+v))$. The figure O-E-T-E is a kite in general and a rhombus exactly when u = v, which is Kulaichev's condition (a); then its diagonals are 1 : u and its angle at O and T is 2 atan u.

yantra	angle O	angle T	angle E	diagonal ratio	shape
Huet classical	78.23°	70.37°	105.70°	0.7552	kite
Kaivalyāśrama grid	75.52°	73.04°	105.72°	0.7572	kite
Kavi Mahesh	78.46°	70.53°	105.50°	0.7579	kite
Kulaichev–Rao–Pustynski	73.77°	73.77°	106.23°	0.75044	rhombus
R1 / Q1 / Q2	76.73 / 78.19 / 77.32°	67.38 / 70.44 / 69.98°	—	0.724 / 0.755 / 0.747	kite

Table 2. DICTO's rhombus in published and new yantras (rhombus.py, rhombus-data.txt).

The 3-4-5 rhombus. Taking u = v = 3/4 (diagonals 3 : 4, inscribed triangles on 9/25, 16/25, 12/25) with (b) and (c) gives an exact yantra, Rh345: S = b_9 = (21464 + 56√56869)/37825 with 56869 = 29·37·53, and Q = b_6 and b_1 have irreducible minimal polynomials of degree 20; all 23 conditions are checked exactly in that number field (rhombus345_exact.py). Kulaichev's condition (d), measured and not imposed, misses by $2.36 \cdot 10^{-5}$ of the diameter; Kulaichev's own yantra has u = 0.750439. The golden rhombus (diagonals 1 : φ) and the 60°/120° rhombus also give admissible exact yantras, far from the classical look (t1's base at 0.65/48 and 0.30/48, incircle off by 0.017 and 0.024); a square is impossible (P = R = 1/2, the bases of t3 and t7 coincide). The natural special rhombus is the 3-4-5 rhombus, not the golden one.

Kulaichev's yantra, Rh345 and Q3 have **45** odd pieces, not 43: their third ring holds 12 pieces, not all of them triangles. This confirms Pustynski's remark (2014, §2.5) about "two additional tiny triangles". Which pairs of slopes bring them is taken up in §9.5.

4. Solids and three dimensions

DICTO proposed the triangles as tetrahedra in a sphere, the stella octangula as the rhombus's solid, and spinning tetrahedra. Seen down its three-fold axis the regular stella octangula is the hexagram; seen from the side, its two tetrahedra show the upright rhombus (Figure 3). Spun about the axis, a regular tetrahedron with a level base sweeps a cone of slope 1/√2 = 0.707, the stella's slope; DICTO's Cleo-Icos (the icosahedron's face pyramid, #29) sweeps a cone of slope $2/\varphi^2$ = 0.764. Huet's classical yantra has 0.813 above and 0.705 below, Kulaichev's 0.750 for both.

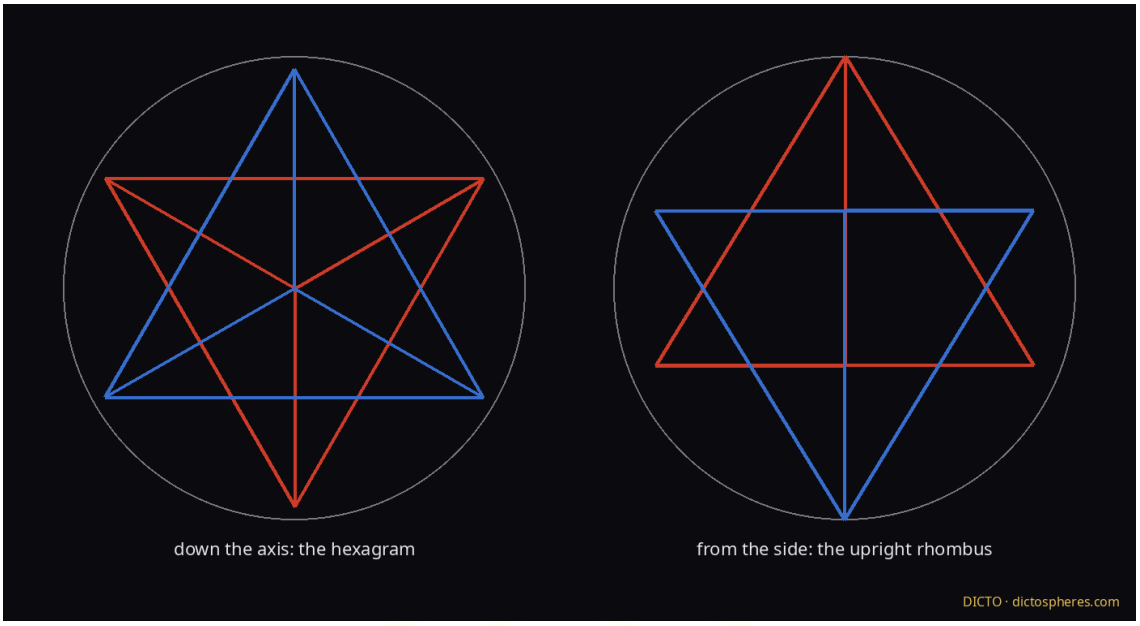


Figure 3. The stella octangula as the yantra's frame. Down the three-fold axis: the hexagram. From the side: the upright rhombus.

> **Proposition 4 (the yantra in three dimensions).** Replace every triangle by a solid with its tip on the axis and a level base. (1) Spun about the axis, each sweeps a cone, and the flat yantra is exactly the section through the axis of the nine nested cones; the 18 marmas become 9 circles, on each of which two cones and one level disc meet. (2) Still, let each be a tetrahedron (triangular pyramid) whose base triangle is inscribed in its cone's base circle, the apex-down ones turned by 60° as in the stella octangula. Seen from each of the three positions on the great circles through the axis that face a base side, the outlines form the flat yantra narrowed sideways by cos 30°, with every meeting kept.

Proof. (1) A triangle with apex on the axis and level base, spun about the axis, gives a cone whose section through the axis is the triangle; so the section of the nine cones is the yantra. A marma at (x, d) lies on two legs and a base; spun, it traces the circle of radius |x| at depth d, which lies on both cones and on the level disc; the mirror point (−x, d) traces the same circle, so 18 points give 9 circles. (2) An equilateral base inscribed in a circle of radius w has side $\sqrt{3}$ w; seen along a direction perpendicular to one side, its outline has half-width $(\sqrt{3}/2)$ w = w cos 30° and the pyramid's outline is the triangle of the same tip and base depth with half-width w cos 30°. Turning a tetrahedron by 60° changes which side faces the viewer but not the outline's width, and the three positions are equivalent by the three-fold symmetry. Scaling every horizontal distance by cos 30° is an affine map, which keeps lines and every meeting. ∎

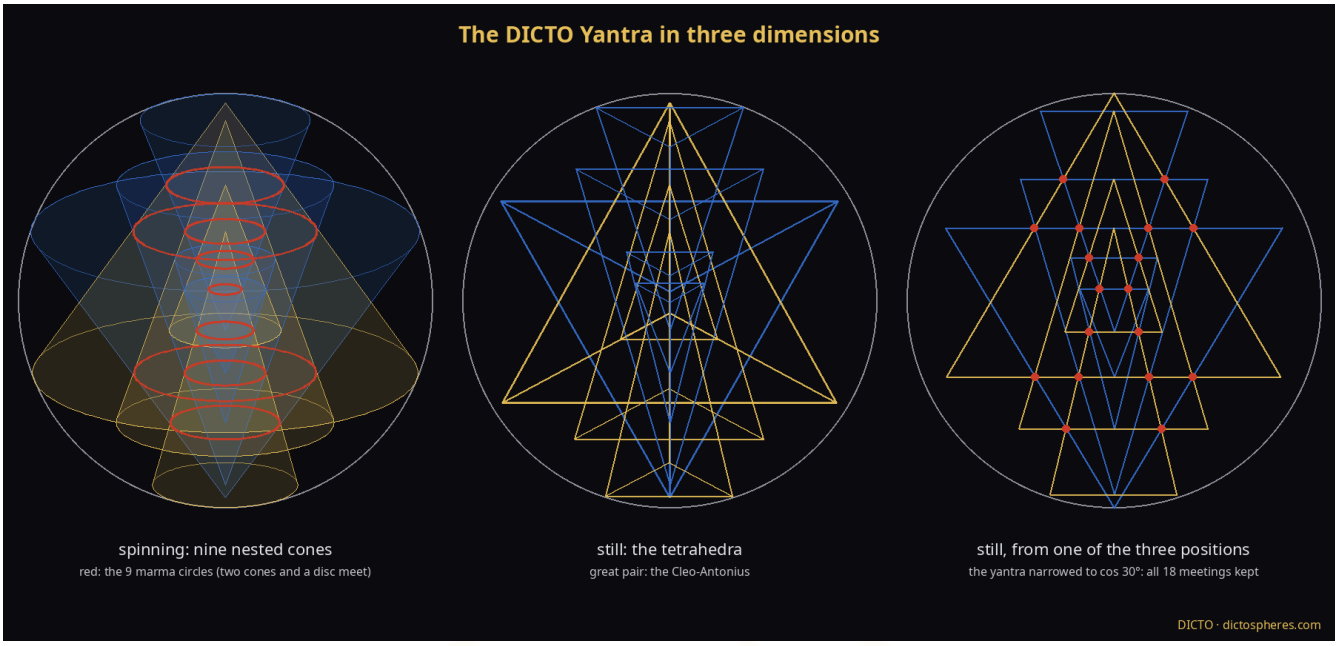


Figure 4. The DICTO Yantra in three dimensions (§9.1). Left: spinning, nine nested cones and the 9 marma circles. Middle: the still tetrahedra, a Cleo-Icos above and a stella tetrahedron below. Right: from one of the three positions, the yantra narrowed to cos 30°, all 18 meetings kept.

5. The viewpoint

DICTO proposed the viewpoint: a change of perspective that makes one yantra meet every demand. A tilted circle seen from the right point still looks round. In the unit disc (y up, O at (0, 1), T at (0, −1)) these perspectives are the maps

$$T_\beta(x, y) = \left(x\sqrt{(1 - \beta^2)} / (1 + \beta y),\ (y + \beta) / (1 + \beta y) \right),\ -1 < \beta < 1,$$

which in modern terms are the translations of Klein's model of the hyperbolic plane along the axis.

> **Theorem 5 (the viewpoint principle).** T_β maps the circle onto itself and keeps the axis, O and T. It keeps every line straight, every meeting and every crossing, every level base level, every corner on the circle, and the order of depths along the axis; hence it keeps (i)–(iii), (b), (c) and the numbers of pieces, marmas and sandhis. It changes the two great slopes to λu and v/λ, with $\lambda = \sqrt{((1 + \beta)/(1 - \beta))}$, so the product uv is unchanged. It does not keep the centre of the bindu's incircle, which is how (d) can be won.

Proof. T_β is a projective map (both coordinates are fractions with the same linear denominator), so lines go to lines and incidences are kept. On the circle, $x^2 = 1 - y^2$ gives $x'^2 + y'^2 = [(1 - y^2)(1 - \beta^2) + (y + \beta)^2]/(1 + \beta y)^2 = (1 + \beta y)^2/(1 + \beta y)^2 = 1$. The axis $x = 0$ is fixed, $(0, \pm 1)$ are fixed, and a level line $y = c$ goes to the level line $y' = (c + \beta)/(1 + \beta c)$; $dy'/dy = (1 - \beta^2)/(1 + \beta y)^2 > 0$ keeps the order of depths. For a point on a leg through O, $1 - y' = (1 - \beta)(1 - y)/(1 + \beta y)$, so $x'/(1 - y') = \lambda \cdot x/(1 - y)$; through T, $1 + y' = (1 + \beta)(1 + y)/(1 + \beta y)$, so $x'/(1 + y') = (x/(1 + y))/\lambda$. The incircle is not a projective notion, so its centre moves. ∎

The Cleo-Octangula yantra K1 (§9.2), seen at $\beta = -0.0405464286$ ($\lambda = 0.9602432222$), has its bindu centred while keeping both small triangles on the circle: every condition at once, with 43 triangles, 18 marmas and 24 sandhis. Its slopes become 0.7335605 above and 0.7363830

below, whose product is still $(2/\varphi^2)(1/\sqrt{2}) = 0.5401815$. This is V1 (Figure 6, right): the perfect yantra of the Cleo-Octangula is the Cleo-Octangula in perspective. A direct solve of "uv fixed, (b), (c), (d)" for the Cleo-Octangula product returns u = 0.733560546791, v = 0.736382996384, the same yantra.

6. Natural coordinates

DICTO proposed a simpler formula. Let t1's corner be the circle point $(x, d) = (t/(1+t^2), t^2/(1+t^2))$ and keep u, v, S.

> **Theorem 6 (natural coordinates).** (1) Condition (c), t9's corners on the circle, is one quadratic in S:
> $G(S) = (u^6v^4 + 3u^4v^4 + u^2v^4 - 2u^2v^2 + 1)\,S^2 - (3u^4v^4 + 2u^2v^4 - 2u^2v^2 + 1)\,S + u^2v^4 = 0.$
> (2) Given (b) through t, the depth Q is rational in t:
> $Q = t\,v\,(u^2t^3 - u^2v\,t^2 - u^2v^2\,t + v) / (u^2v\,t^4 - u^2(v^2+1)\,t^3 - u^2v(v^2-1)\,t^2 + (v^2+1)\,t - u^2v^3).$
> (3) The construction's closing equation becomes one relation H(t, S) = 0 whose only non-trivial factor has degree 10 in t and 3 in S (the other factors are u, v, tu ± 1 and t ± v).

Proof. natural.py derives G, Q and H symbolically from the model: (c) is $w_9^2 = S(1 - S)$ with w_9 linear in S; the marma (1,3,8) makes Q linear once t1's leg is fixed by t; the closing (Apollonius) equation of the construction is then cleared of denominators and factored. On the DICTO Yantra the three residuals are $1.7\cdot10^{-32}$, $8.8\cdot10^{-34}$ and $3.6\cdot10^{-31}$. ∎

The earlier blind searches for exact forms failed because the old coordinates hid this structure. With u and v fixed by solids, (b) and (c) leave a curve given by G and H; (d) is one more equation, and dicto_exact.py eliminates S and t from G, H and the bindu condition to reach one polynomial for u over $\mathbb{Q}(\sqrt{5})$. It always terminates; it was still running at release (8 November 2026), and its result will be added in a later version.

7. One slope for all

DICTO proposed the triangular grid, after the Matos painting, and then the Cleo grid: all nine triangles similar, with one leg slope s (s = 1/√3 equilateral, s = 1/φ the Cleo-Icos side face).

> **Theorem 7 (one slope).** If all nine triangles have the same leg slope s, then: (1) the 18 marmas and the 7 apex-on-base conditions fix the yantra up to scale and shift along the axis, for every s, with heights t5 : t1 : t4 : t6 : t8 : t9 : t2 : t3 : t7 = 2 : 4 : 4 : 4 : 6 : 6 : 8 : 8 : 9; (2) adding the corner touches forces every triangle to size zero, so no one-slope yantra has them; (3) the two great triangles cannot both be inscribed.

Proof. With one slope, two legs of slope s from tips at depths a_i and a_k, one pointing down and one up, cross at depth $(a_i + a_k)/2$, whatever s. So a marma (i, j, k) is the linear condition $b_j = (a_i + a_k)/2$, an apex-on-base is $a_x = b_y$, and a corner touch of i on j is $b_i = (a_i + a_j)/2$. Solving the 16 conditions of (1) leaves two free depths and gives the sizes $a_i - b_i$ as multiples of $b_9 - b_8$: 2, 4, 4, 2, 1, 2, 9/2, 3, 3 for t1 … t9, which doubled are the ratios above. Adding the three corner touches leaves one free depth and every size 0 (one_slope.py). (3) An inscribed triangle with its tip on the circle and leg slope s has height $1/(1 + s^2)$; t3 and t7 would need the same height, but their heights are in the ratio 8 : 9. ∎

With t7 inscribed, t3's tip falls outside the circle, at depth 7/6 = 1.167 for the equilateral grid and 1.126 for the Cleo grid, and the pieces are 12, 9, 6, 9, 1 (37), not 43. The slope only stretches the yantra sideways, so the two grids give the same yantra up to that stretch. A grid yantra is exact in every meeting but is not the traditional figure; painted grid yantras relax exactness.

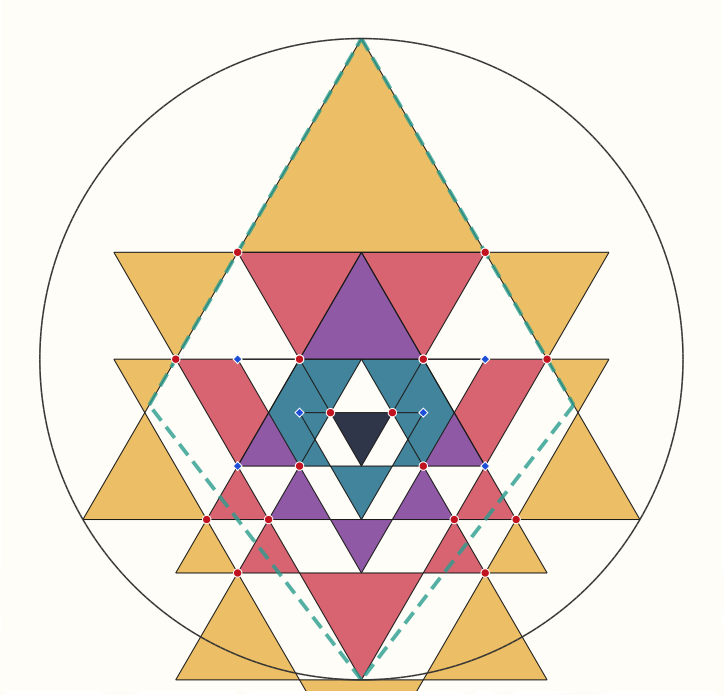


Figure 5. The equilateral grid yantra (Theorem 7): every meeting exact, heights 2 : 4 : 4 : 4 : 6 : 6 : 8 : 8 : 9; t3's tip lies outside the circle and the odd pieces number 37 (12, 9, 6, 9, 1).

8. The parent of the perfect stella yantra

With both great slopes fixed by solids, only two of (b), (c), (d) can be imposed. DICTO chose the stella: keep the stella's tetrahedron exact below ($v = 1/\sqrt{2}$) and free the upper slope. The result, PF2 (§9.3), has upper slope $u = 0.695886431693$, whose base corners lie 110.332934° round the circle from its tip ($\cos = (u^2 - 1)/(u^2 + 1)$); the stella's lower triangle has 109.471221°, the tetrahedral angle.

> **Proposition 8 (no parent solid).** (1) No two corners of a Platonic solid (Dodecahedron, Icosahedron, Cube, Tetrahedron, Octahedron), of a regular 4D polytope (5-cell, tesseract, 16-cell, 24-cell, 600-cell, 120-cell), or of any of the 33 solids of DICTO's dataset v8 (of 102) whose corners lie on one sphere, subtend 110.332934° at the centre; the nearest are 0.064° away (Lesser Golden Cut Snub Dodecahedron) and, among regular solids, 0.121° (120-cell). (2) In dimension 5 and higher, the regular polytopes (simplex, cube, cross-polytope) and the exceptional polytopes of E6, E7, E8 have rational corner coordinates, so every cosine between their corners is rational; none can hold the angle if u^2 is irrational.

Proof. (1) is the finite search search.py (data-search.txt), which lists every angle between corners of every listed solid. (2) $\cos \alpha = (u^2 - 1)/(u^2 + 1)$ is rational exactly when u^2 is. findpoly finds no integer polynomial of degree ≤ 8 for u, which supports, but does not prove, that u^2 is irrational. Golden symmetry stops at H4 in dimension 4, which (1) covers. ∎

So the upper slope of the perfect stella yantra is the yantra's own number. Regular polygons with many sides (angles a rational fraction of a turn) were not searched.

9. Computed yantras

> Every yantra in this section lies in the family of Theorem 1, so its 18 marmas, corner touches and apex-on-base points are exact for its parameters. What is **computed** is which parameters meet the further conditions: solved with 60-digit arithmetic to 10^{-50} (E1: double precision, then rebuilt at 60 digits); the values below are cut from the stored files without rounding. Exact algebraic forms are open.

yantra (great pair)	u (upper)	v (lower)	(b) (c)	(d)	pieces	marmas	sandhis	nesting k
DICTO Yantra (GF1) Cleo-Antonius, from a viewpoint	0.6793185887	0.6950140374	yes	yes	43	18	24	0.05498385
Cleo-Octangula K1 Cleo-Icos / stella	$2/\varphi^2$ = 0.763932	$1/\sqrt{2}$ = 0.707107	yes	no	43	18	24	—
Cleo-Octangula V1 Cleo-Octangula, from a viewpoint	0.7335605468	0.7363829964	yes	yes	43	18	24	0.05750553
perfect stella yantra PF2 free / stella	0.6958864317	$1/\sqrt{2}$	yes	yes	43	18	24	0.05634293
PF1 Cleo-Icos / free	$2/\varphi^2$	0.7621139499	yes	yes	45	18	28	0.05617825
gnomon yantra G1 Cleo-Icos / stella	$2/\varphi^2$	$1/\sqrt{2}$	(b) only	yes	43	18	24	0.06307419
K2 stella / Cleo-Icos	$1/\sqrt{2}$	$2/\varphi^2$	yes	no	43	18	24	—
K3, K4 $\varphi/2$ / stella, $\varphi/2$ / Cleo-Icos	$\varphi/2$	$1/\sqrt{2}$, $2/\varphi^2$	yes	no	45	18	28	—
stella rhombus yantra stella / stella	$1/\sqrt{2}$	$1/\sqrt{2}$	yes	no	43	18	24	—
Cleo rhombus yantra Cleo-Icos / Cleo-Icos	$2/\varphi^2$	$2/\varphi^2$	yes	no	45	18	28	—
Kulaichev–Rao–Pustynski	0.7504385160	0.7504385160	yes	yes	45	18	28	—
Rh345 3-4-5 rhombus	3/4	3/4	yes	no	45	18	28	—
most even E1	0.7777407207	0.7434588452	no	not imposed	43	18	28	—

Table 3. The computed yantras (golden_family.json, kites.json, viewpoint.json, perfect.json, gnomon.json, even.json, cleo-yantra.json, kulaichev-numeric-data.txt). Pieces = odd pieces (figure.py); marmas and sandhis by counts.py with exact fractions from the stored digits; for E1, Kulaichev's yantra, Rh345, the stella and Cleo rhombus yantras and K1 the counts were redone for this paper at 60 digits; k = bindu inradius / circle radius (§9.4).

9.1 The DICTO Yantra

DICTO proposed a side of Antonius in place of the stella. Antonius (D65, the golden antiprism, #44) is the Cleo-Icos's own partner: Cleo + Antonius + Cleo make the golden rhombohedron. Its side faces have slope exactly $1/\varphi$ (apex 63.43°). DICTO named the pair the **Cleo-Antonius**: the Cleo-Icos cone above ($2/\varphi^2$), the Antonius side below ($1/\varphi$), product $2/\varphi^3$, all in $\mathbb{Q}(\sqrt{5})$. By Theorem 5 only the product matters for (b), (c) and the counts, and a viewpoint can

then win (d). The perfect member, solved directly from "uv = $2/\varphi^3$, (b), (c), (d)", is the **DICTO Yantra** (Figure 1):

$$u = 0.679318588706774657388650310391,$$
$$v = 0.695014037372934486335281853716,$$

with P = 0.3257114053, Q = 0.5765103217, R = 0.6842408243, S = 0.9682686075, every condition met, 43 triangles, 18 marmas and 24 sandhis. Its product uv equals $2/\varphi^3$ = 0.4721359550. Dividing u by $2/\varphi^2$ gives λ = 0.8892395772, and $v \cdot \lambda = 1/\varphi$: by Theorem 5 the DICTO Yantra is, exactly, the yantra on the Cleo-Icos and the Antonius side (with (b) and (c)) seen through T_β with $\beta = (\lambda^2 - 1)/(\lambda^2 + 1) = -0.11685$. DICTO named it the DICTO Yantra.

9.2 The Cleo-Octangula

DICTO proposed the Cleo-Icos as the great pair, then the kite proportion: keep the Cleo but not the rhombus. The **Cleo-Octangula** is DICTO's name for the pair of a Cleo-Icos above and the stella's tetrahedron below ($u = 2/\varphi^2$, $v = 1/\sqrt{2}$). With (b) and (c) it gives K1: valid, exactly the traditional 43 triangles (14, 10, 10, 8, 1), 18 marmas, 24 sandhis; P = 1/3 exactly (the lower great triangle's base at one third), Q = 0.53625 (Huet's classical 0.537), R = 0.63148, S = 0.93007. Its numbers need both $\sqrt{5}$ (the Cleo) and $\sqrt{2}$ (the stella). It misses (d), and its perfect form is V1 (§5).

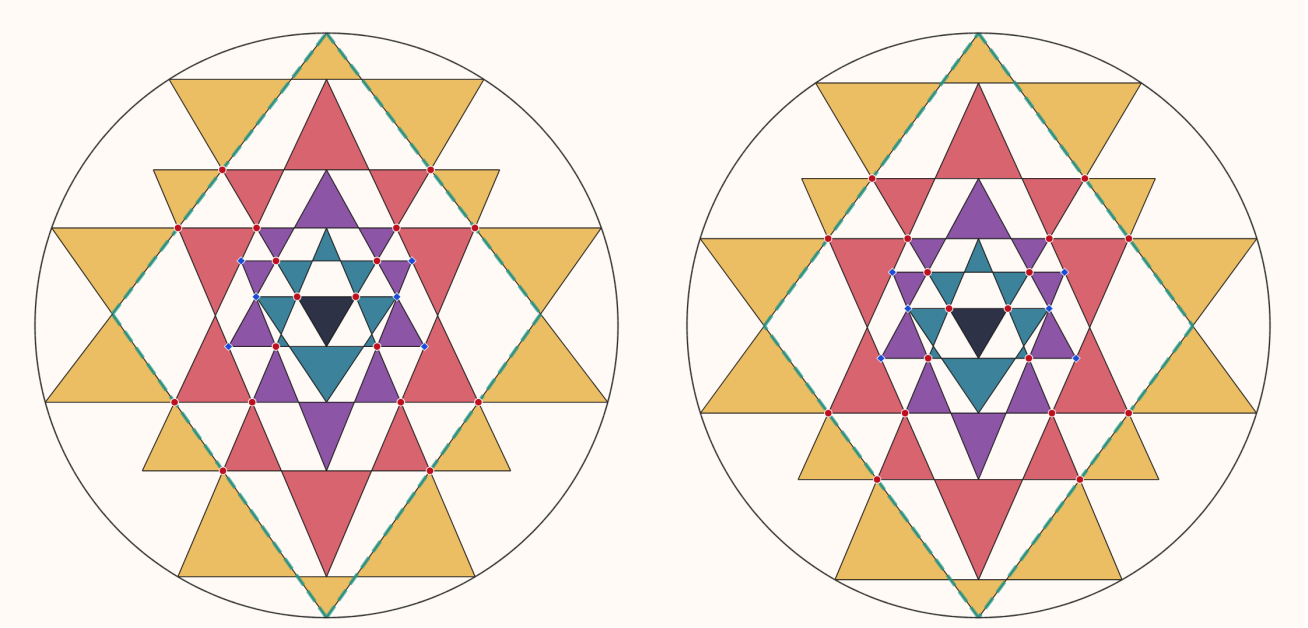


Figure 6. Left: the Cleo-Octangula yantra K1, a Cleo-Icos above and the stella's tetrahedron below, both small triangles on the circle. **Right: V1**, the same yantra from the viewpoint that centres the bindu (Theorem 5): every condition exact.

9.3 The perfect stella yantra

Fixing both great slopes leaves room for only two of (b), (c), (d). DICTO chose the stella: PF2 keeps the stella's tetrahedron exactly below and frees the upper slope, 0.695886 (apex 69.67° against the stella's 70.53°). Then every condition holds at once, with the traditional 4 up, 5 down, 43 triangles, 18 marmas and 24 sandhis (Figure 8, left). Its upper slope has no parent solid (Proposition 8). Keeping the Cleo-Icos above instead (PF1) also meets (b), (c), (d), but has 45 pieces and 28 sandhis.

9.4 The gnomon

DICTO proposed a gnomon like his Julias (#54). When the bindu's incircle is centred, a copy of the whole yantra scaled by k = (incircle radius)/(circle radius) fits that incircle exactly and concentrically; the yantra outside it is a gnomon and the yantras nest without end. Every yantra meeting (d) nests: the DICTO Yantra at k = 0.0549838452, V1 at 0.0575055271, PF2 at 0.0563429292. The figure shows G1, the Cleo-Octangula with (b) and (d) only (k = 0.0630741873, 1/k = 15.8543; t9's corners 0.066 inside the circle); the Cleo-Octangula with (c) and (d) gives no valid yantra.

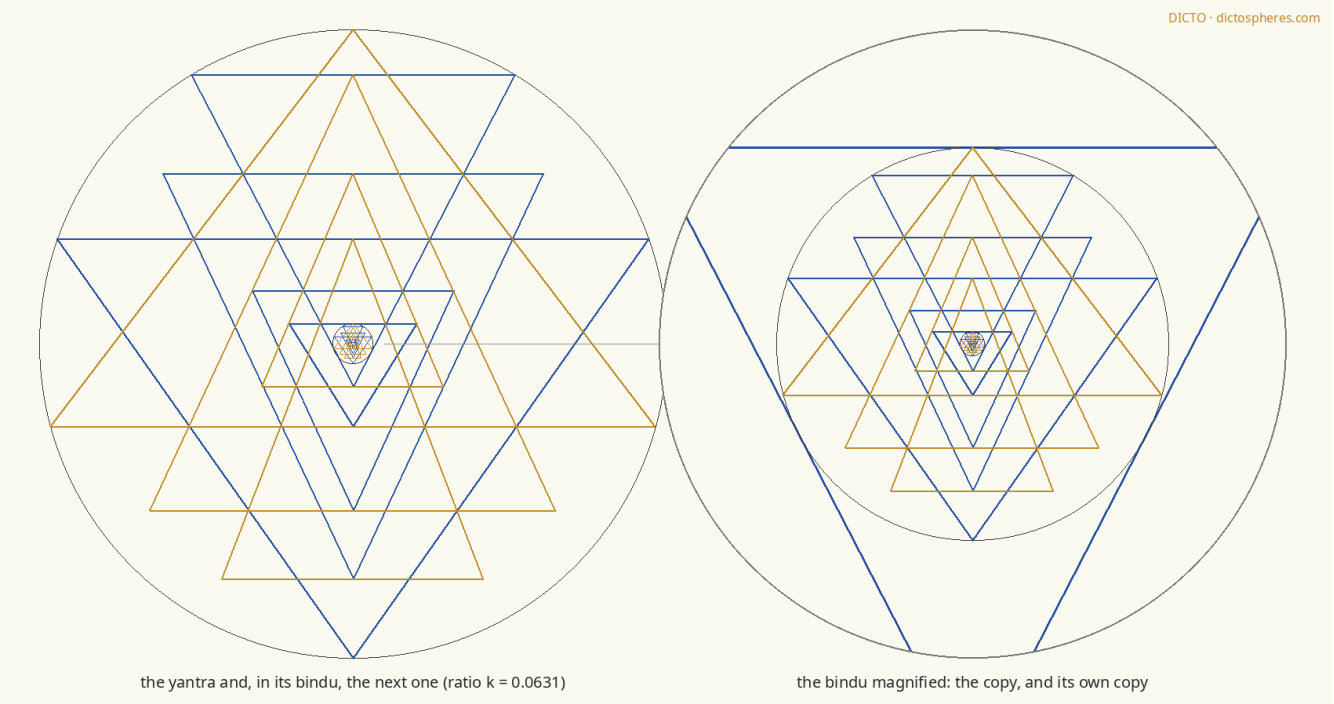


Figure 7. The gnomon (G1). Left: the yantra and, in its bindu, the next one. Right: the bindu magnified; the copy touches the bindu triangle's three sides, and holds its own copy.

9.5 Equal great triangles and the two extra pieces

The two extra tiny pieces (45 odd pieces with 28 sandhis instead of 43 with 24) appear in every one of Kulaichev's yantra, Rh345, Q3 and the Cleo rhombus yantra, all with u = v; but the stella rhombus yantra (u = v = 1/√2, with (b) and (c)) has the traditional 43 and 24, and the unequal pairs K3, K4 and PF1 have 45 and 28. So equal great triangles do not decide the count: it depends on the slopes themselves. In every yantra of Table 3 with (b) and (c), 45 pieces came with 28 sandhis and 43 pieces with 24. A rule for when the extra pieces appear is open.

9.6 The most even exact yantra

Among exact yantras (all of Theorem 1's conditions, great triangles inscribed) with the traditional 43 pieces, E1 minimises the spread of the 43 areas (standard deviation over mean): u = 0.7777407207, v = 0.7434588452, Q = 0.5415589952, S = 0.8719616408, spread/mean 0.614, against 0.677 for Huet's classical yantra and 0.866 for the DICTO Yantra. Its t1 and t9 corners lie 0.037 and 0.047 inside the circle, as in painted yantras. The pieces

must shrink ring by ring toward the bindu: the smallest is 0.020 of the largest. E1 keeps 43 pieces and 18 marmas but has 28 two-line crossings.

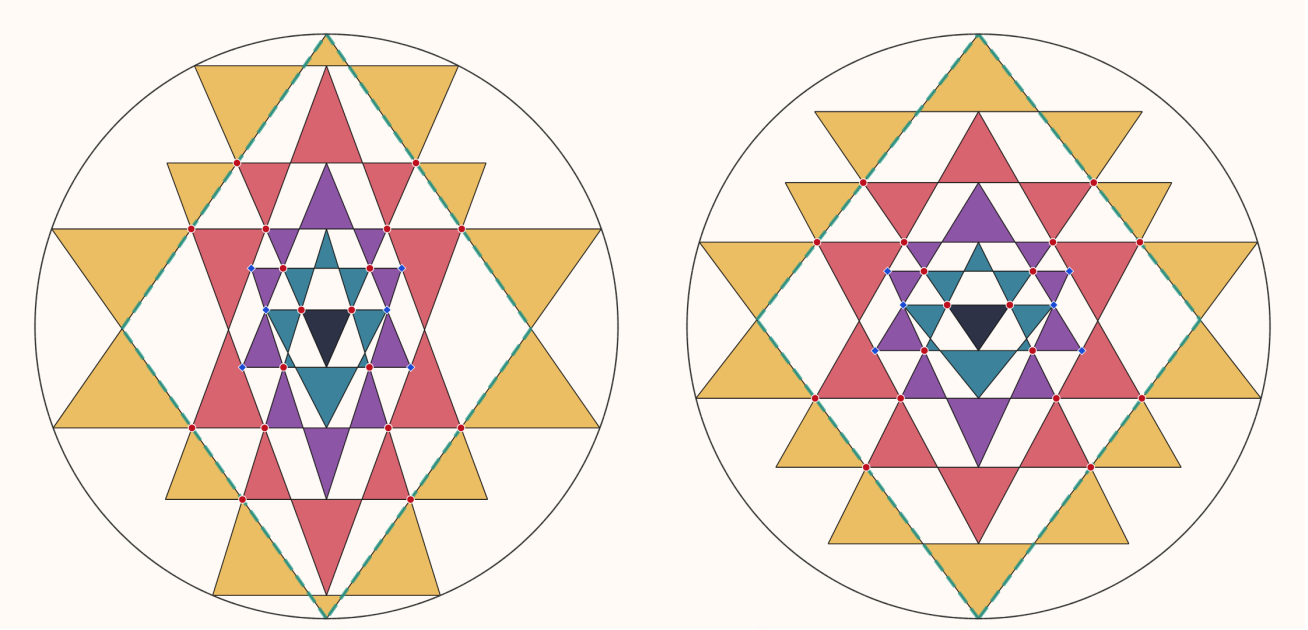


Figure 8. Left: the perfect stella yantra PF2, the stella's tetrahedron exact below, every condition exact. **Right: the most even exact yantra E1**: 43 pieces as equal as the structure allows, its small triangles just inside the circle.

10. Spheres

Projected from the centre of a sphere onto a plane touching it (gnomonic projection), every line of a yantra becomes a great circle and every meeting is kept (each marma is three planes through the centre sharing one line); the yantra's circle becomes a small circle, and every exact condition holds for any cap size. The upper great triangle cannot be made an exact spherical icosahedron face: its tip angle starts at 72.53° in the flat perfect yantra V1 and only grows on the sphere, and its base angles never reach equality (cleo_cap.py).

DICTO proposed the Cleo sphere: twenty yantras (V1), one in the inscribed circle (angular radius 20.9052°) of each face of the spherical icosahedron, that is, on the base of each of the twenty Cleo-Icos; adjacent caps touch and twelve gaps sit at the icosahedron's corners. DICTO then proposed the 81-triangle sphere: each of the twenty faces split into four at its side midpoints, giving 80 triangles (20 central, 60 at the corners), 42 corners and 120 edges (Euler characteristic 2), with a DICTO Yantra in each triangle's inscribed circle and the corner yantras pointing to their icosahedron corner (five-petalled flowers at the 12 corners). With the bindu at the centre the count is 81, the number of figures of the Kongōkai mandala of 81 deities (Kongōkai hachijūisson mandara), the rarer form beside the usual Nine Assemblies (Kue) mandala, brought from China by Ennin and used mainly in Tendai esoteric Buddhism [12].

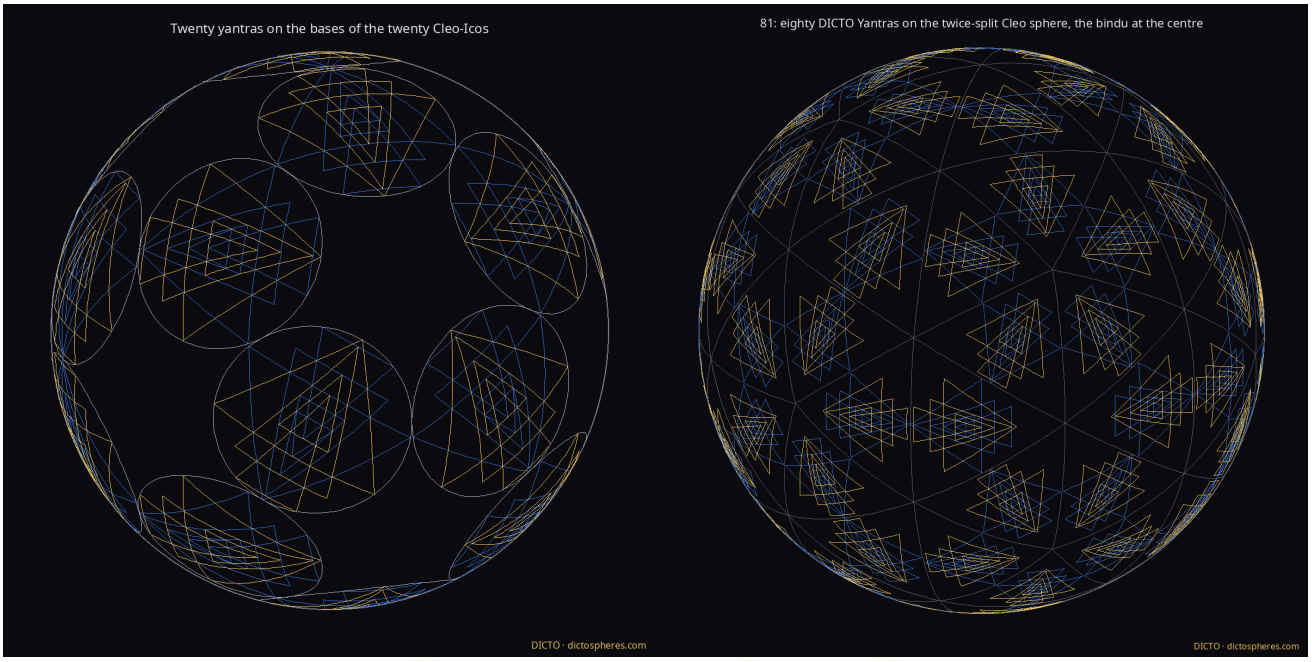


Figure 9. Left: the Cleo sphere, twenty DICTO Yantras on the bases of the twenty Cleo-Icos. **Right: the 81-triangle sphere**, eighty DICTO Yantras on the Cleo sphere split once more, the bindu at the centre.

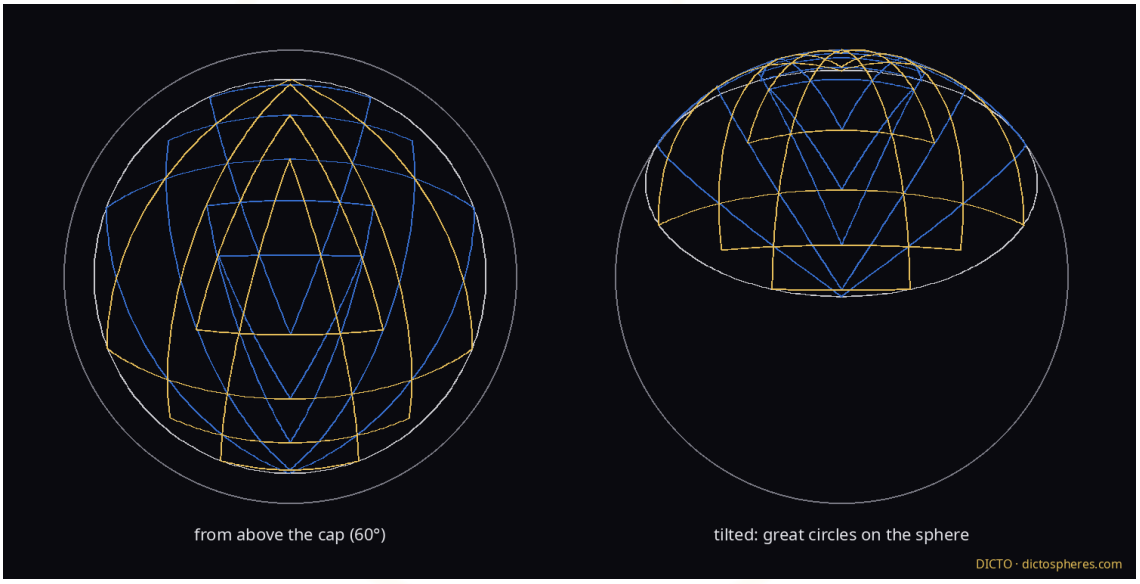


Figure 10. The spherical DICTO Yantra (on a 60° cap): lines become great circles, every meeting kept; seen from above the cap and tilted.

11. The choice

Every exact yantra found chooses between two goods: its small top and bottom triangles touching the circle, or its 43 pieces as even as they can be. The **DICTO Yantra** takes the first: on the Cleo-Antonius, every traditional condition holds, the corners touch the circle, the bindu is centred, 43, 18 and 24; it is perfect by the rules. The **most even yantra** takes the second: every meeting and the 43 kept, its corners just inside the circle as in painted yantras; it is perfect to the eye. The grid yantras keep every meeting with integer heights but lose both: 37 pieces, and the circle cannot hold them.

12. What is open

(1) Exact algebraic forms of the DICTO Yantra, V1, PF2 and E1 (the elimination of §6 is the route). (2) Kulaichev's yantra exactly: the resultant run (kulaichev_exact.py) reached $\mathrm{Res}_t(\mathrm{core}, E_b)$ and $\mathrm{Res}_t(\mathrm{core}, E_d)$, of degrees (56, 10, 12) and (54, 11, 12) in (u, Q, S), and stopped at the next step within its 20-minute limit; no integer polynomial of degree ≤ 96 for u, Q or S at 3000 digits; Kulaichev bounded the degree by 12544. (3) When the two extra pieces appear (§9.5). (4) A proof that the evenness optimum E1 is global, and that no yantra with corners on the circle is as even. (5) A rational yantra nearer the traditional drawing (rational points on the genus-2 fibres); a rational yantra on the rhombus (Δ a square on a three-parameter slice). (6) Kavi Mahesh's rational heights made exact, in $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{\Delta})$. (7) Irrationality of u^2 for PF2 (Proposition 8 (2)).

13. What is known, and what is new

Known, credited. (Huet 2002 and Kulaichev 1984 are known here through Chiodo, Pustynski and Kavi Mahesh, whose numbers our model reproduces; their full texts were not available to us.) Huet (1980s, published 2002) showed that the yantra has four free real parameters (five starting values tied by one equation) and solved it by Newton iteration; his classical yantra has P, Q, R, S = 0.332, 0.537, 0.602, 0.835 in Chiodo's form. Chiodo (2021) proved the whole four-parameter family constructible by straightedge and compass, the closing step being Apollonius' circle–line–point problem; so constructible parameters give constructible coordinates. Chiodo gives no explicit field, minimal polynomial or exact example. Kulaichev (1984), Rao (1998) and Pustynski (2014) fixed one rigid yantra by four extra conditions, known to 15 figures (Pustynski's Table 1, which our model reproduces to $3.9 \cdot 10^{-15}$, all 14 of his significant figures); Kulaichev bounded its degree by 12544, and Pustynski thought a purely analytical solution of the last two steps "probably impossible" and remarked on two additional tiny triangles. Kavi Mahesh (2023) used rational heights with 16 of the 18 marmas exact by construction and two missing by 0.02% of the diameter. Tripathi (2025) drew it numerically with residual errors. Gonçalves, Mádr and Stříž (2009) drew the core in MetaPost. Kaivalyāśrama's traditional grid (P, Q, R, S = 17, 27, 30, 42 out of 48, as Chiodo suggests for traditional diagrams) puts t1's base corners outside the circle (power +0.031): the grid itself is not admissible.

New here, as far as these sources go. The quadratic family with an explicit field (Theorem 1); a rational yantra (Theorem 2); the 43 as the odd pieces (Proposition 3); DICTO's rhombus and the exact 3-4-5 rhombus yantra; the yantra on solids (the stella octangula, the Cleo-Icos, the Cleo-Octangula, the Cleo-Antonius), in three dimensions (Proposition 4) and on spheres; the viewpoint principle (Theorem 5) and the perfect yantras it gives, the DICTO Yantra first; natural coordinates (Theorem 6); the one-slope theorem (Theorem 7); the gnomon nesting; the most even yantra and the choice of §11. We found none of these in the sources below, and claim no more than that.

Sources checked (8 November 2026). G. Huet, Śrī Yantra Geometry (2002): full text not reachable; used as quoted by Chiodo 2021 and Kavi Mahesh 2023. A. Chiodo, On the construction of the Śrī Yantra (2021): full text read. V.-V. Pustynski, On mathematical complexity of Śrīyantra (2014): full text read from insa.nic.in/writereaddata/UpLoadedFiles/IJHS/Vol49_3_5_Pustynski.pdf (the link DICTO gave, cahc.jainuniversity.ac.in, returns 404). A. P. Kulaichev, Śrīyantra and its Mathematical Properties (1984): the INSA PDF is a scan with no text layer; used as summarised by Pustynski and Chiodo. Kavi Mahesh, Understanding the geometry of Sri Chakra (2023): full text read at anantaajournal.com/archives/2023/vol9issue6/PartD/9-6-39-715.pdf. L. N. Gonçalves, M. Mádr, P. Stříž, Drawing

the Shree Yantra Core with Metapost (2009): abstract page only (DML-CZ 10338.dmlcz/150096). B. Tripathi, texample.net/sri-yantra (TikZ, 30 November 2025). Web searches: "Pustynski Sri Yantra …", "Huet Sri Yantra Geometry … parameters", "arXiv Sri Yantra exact construction concurrency algebraic": no arXiv paper found. Named in Chiodo's bibliography and not read: Bolton–Macleod 1977, Fonseca 1986, C. S. Rao 1998, Tahta 1992, Santhi et al. 2012, Tularam 2012. Reference library on dicto-big-brother (~/kaleidohedra-references, every MANIFEST.json and text file) grepped for "yantra" and "Huet": no hits. H. J. T. Matos, painted Śrī Yantra (2023, CC BY-SA 4.0): DICTO's reference image, not reproduced.

Credits

DICTO proposed the upright rhombus; DICTO proposed the triangles as tetrahedra in a sphere; DICTO proposed the stella octangula as the rhombus's solid; DICTO proposed spinning tetrahedra; DICTO proposed the Cleo-Icos as the great pair; DICTO proposed the kite proportion; DICTO proposed a gnomon like his Julias; DICTO proposed the viewpoint; DICTO proposed a side of Antonius in place of the stella; DICTO proposed a simpler formula (natural coordinates); DICTO proposed the triangular and Cleo grids; DICTO proposed the spherical yantra, the Cleo sphere and the 81-triangle sphere. DICTO chose the stella for the perfect stella yantra. DICTO named the DICTO Yantra, the Cleo-Antonius and the Cleo-Octangula; the Cleo-Icos (#29) and Antonius (D65, #44) are DICTO's earlier pieces. DICTO's reference images: his photograph of a gold embossed yantra, and H. J. T. Matos's painting (CC BY-SA 4.0), used as a reference only and not reproduced. Computation and drafting with Claude (Anthropic).

Data and code

Study 2026-11-08-sri-yantra: model.py (shared model; reproduces Pustynski's Table 1 to $3.9{\cdot}10^{-15}$), quadfield.py and quadratic_family.py (Theorem 1, exact $\mathbb{Q}(\sqrt{\Delta})$ arithmetic), discriminant.py, rational_search.py and rational_search2.py (Theorem 2 searches), check_r1_independent.py, rhombus.py and rhombus345_exact.py (§3), figure.py and counts.py (Proposition 3, Table 3), stella_yantra.py, cleo_yantra.py, kites.py, gnomon_yantra.py and render_gnomon.py, perfect_yantra.py, viewpoint.py (Theorem 5), golden_family.py (the DICTO Yantra), natural.py and dicto_exact.py (Theorem 6), lattice_yantra.py, lattice_clash.py, one_slope.py and grid_yantra.py (Theorem 7), even_yantra.py (E1), kulaichev_numeric.py, kulaichev_exact.py and kulaichev_algdep.py. Study 2026-11-08-tetra-yantra: frame.py, render3d.py (Proposition 4), spherical.py, cleo_cap.py, cleo_sphere.py, sphere81.py. Study 2026-11-08-yantra-4d: search.py, data-search.txt (Proposition 8). Heavy runs on dicto-big-brother; each script ends with a checked FINAL line (grid_yantra.py prints FINAL FAIL by design: t3's tip lies outside the circle, which is Theorem 7's finding).

References

1. G. Huet, Śrī Yantra Geometry, Theoretical Computer Science 281 (2002) 609–628.
2. A. Chiodo, On the construction of the Śrī Yantra, Comptes Rendus Mathématique 359 (2021) 377–397, doi:10.5802/crmath.163.
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7. B. Tripathi, Sri Yantra, texample.net/sri-yantra (30 November 2025).
8. H. J. T. Matos, painted Śrī Yantra (2023), CC BY-SA 4.0 (reference image, not reproduced).
9. DICTO, Theaetetus (#29), DICTO DOI: 10.5281/zenodo.23378070 (the Cleo-Icos).
10. DICTO, Aristotle's Pyramid (#44), DICTO DOI: 10.5281/zenodo.23529497 (Antonius, D65).
11. DICTO, One Turn Is a Gnomon (#54), DICTO DOI: 10.5281/zenodo.23633365 (the Julias).

12. JAANUS (Japanese Architecture and Art Net Users System), "Kongōkai hachijūisson mandara 金剛界八十一尊曼荼羅", projects.mcah.columbia.edu/jaanus/node/2878 (consulted 8 November 2026).